

Proof (Back-door Criterion):

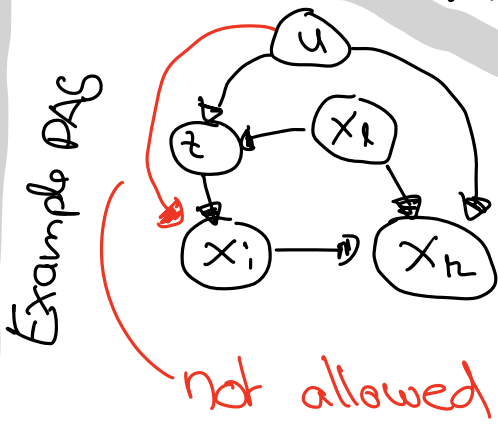
Conditions:

- X_i, X_r are ordered w.r.t. DAG $\mathcal{G}$

- Z satisfies back door if

- i) no node in Z is in $De(X_i)$

- ii) Z blocks all paths between X_i and X_r with an arrow into X_i



Parent adjustment

$$p(x_r | do(x_i = x_i)) = \int p(x_r | x_i, x_{Pa(i)}) p(x_{Pa(i)}) dx_{Pa(i)}$$

Introduce another set of conditioning variables Z :

$$= \int \left[\int \underbrace{p(x_r, x_z | x_i, x_{Pa(i)})}_{\text{Rewrite as } p(x_r | x_z, x_i, x_{Pa(i)}) p(x_z | x_i, x_{Pa(i)})} dx_z \right] p(x_{Pa(i)}) dx_{Pa(i)}$$

Rewrite as $p(x_r | x_z, x_i, x_{Pa(i)}) p(x_z | x_i, x_{Pa(i)})$

$$= \int \left[\int p(x_r | x_z, x_i, x_{Pa(i)}) p(x_z | x_i, x_{Pa(i)}) dx_z \right] p(x_{Pa(i)}) dx_{Pa(i)}$$

Assume that Z satisfies the back-door criterion:

By (ii) " Z blocks all back-door paths": $x_r \perp\!\!\!\perp x_{Pa(i)} | \{x_i, x_z\}$

$$\Rightarrow \int \left[\int p(x_z | x_z, x_i, \cancel{x_{pa(i)}}) p(x_z | \cancel{x_i}, x_{pa(i)}) dx_z \right] p(x_{pa(i)}) dx_{pa(i)}$$

By (i) "no node in z is in $De(x_i)$ " : $\underbrace{x_i \perp\!\!\!\perp x_z | x_{pa(i)}}_{\text{local Markov}}$

Thus: $p(x_z | do(X_i = x_i))$

$$= \int \left[\int p(x_z | x_z, x_i) p(x_z | x_{pa(i)}) dx_z \right] p(x_{pa(i)}) dx_{pa(i)}$$

Since $p(x_z) = \int p(x_z | x_{pa(i)}) p(x_{pa(i)}) dx_{pa(i)}$

$$p(x_z | do(X_i = x_i)) = \int p(x_z | x_i, x_z) p(x_z) dx_z \quad \square$$